

range of services and environments. A true open spectrum environment would allow the same degree of openness, flexibility, and scalability for communication that the Internet itself promotes for applications and content.

There are two ways to implement open spectrum technologies. The first is to designate specific bands for unlicensed devices, with general rules to foster coexistence among users. This is the approach that allowed WiFi to flourish in the 2.4 GHz and 5 GHz bands. The second mechanism is to “underlay” unlicensed technologies in existing bands, without disturbing licensed uses. This approach, epitomized by the ultra-wideband technology the FCC authorized earlier this year, effectively manufactures new capacity by increasing spectrum efficiency. Underlay can be achieved either by using an extremely weak signal or by employing agile radios able to identify and move around competing transmissions.

Both unlicensed bands and underlay have their place. Eventually, underlay approaches will be more significant, because they can work across the entire spectrum rather than requiring the creation of designated “parks.” Someday, if underlay is successful enough, we may not need licensed bands at all, but that day is well in the future. The important point today is to allow both unlicensed bands and underlay to develop based on technological capabilities and market demand. That involves four steps: removing limitations in existing rules, creating additional unlicensed bands, establishing rules to facilitate additional forms of underlay, and funding research into next-generation technologies.

By Land Or By Sea

Open spectrum is actually a simple concept. It requires no flights of fancy about the laws of physics. It sounds strange because, as the examples above suggest, we are accustomed to thinking of the radio spectrum as a scarce physical entity, like land. Charts showing the partitioning of the spectrum and auctions for geo-